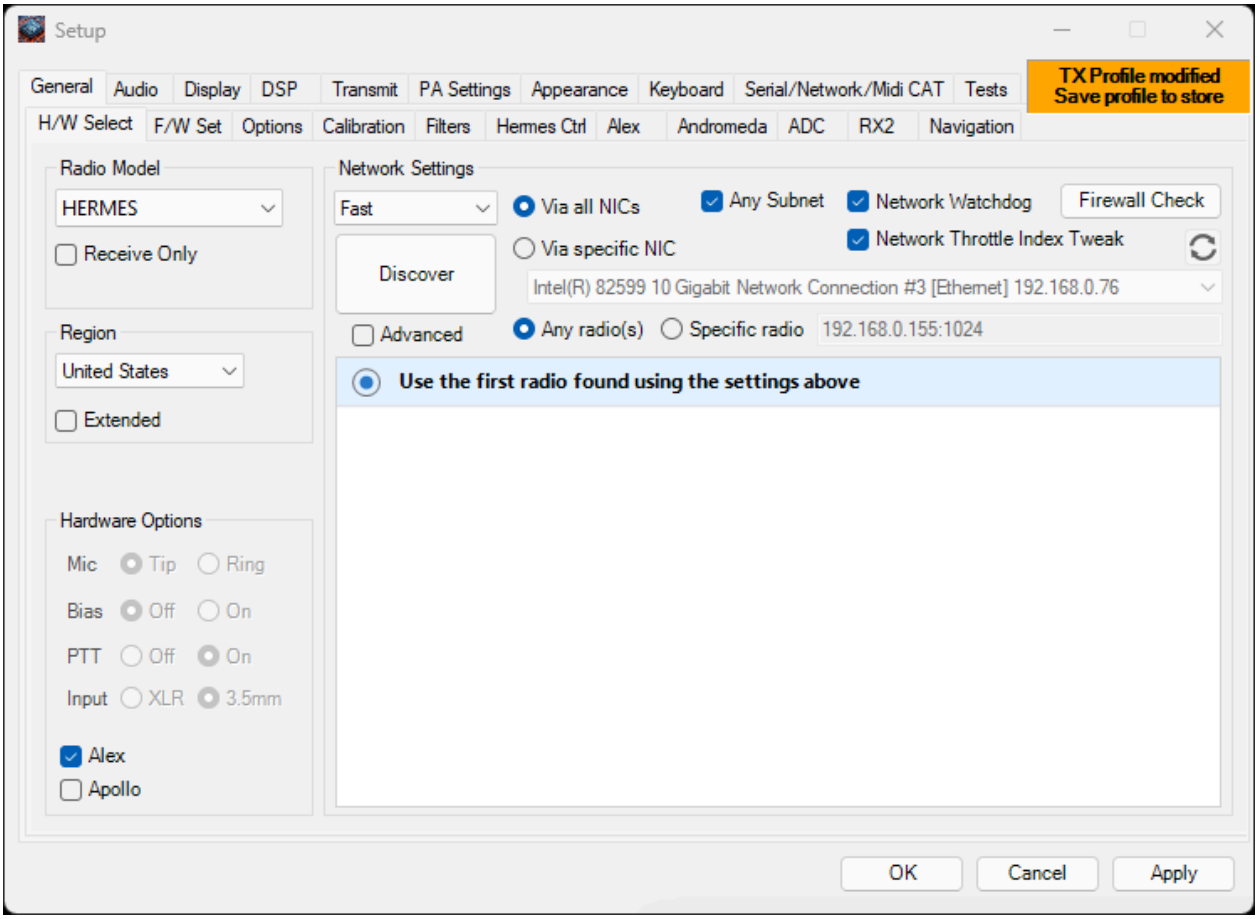
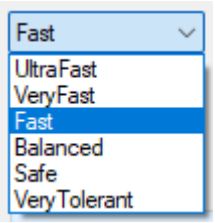
Thetis Network Settings - v2.10.3.13



Discovery and Radio Management

The networking settings now control how Thetis discovers radios on the network, how radios are added to the radio list, and how connections are made to those radios. Discovery occurs when the Discover button is pressed allowing a user to add a discovered radio directly. A discovery is also attempted when Thetis connects to a radio already added to the list of available radios. If a custom radio is added manually, then a discovery is not made and can help in some situations.

Discovery Speed

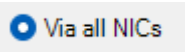


The discovery speed controls how long Thetis waits for radios to respond during a discovery operation. Faster modes use shorter timeouts and fewer retries, while slower modes allow additional time for responses.

On a local LAN, UltraFast or VeryFast can usually be used. Fast is the default setting and provides a good balance, while slower options extend timeouts and retry attempts for less reliable or higher-latency networks.

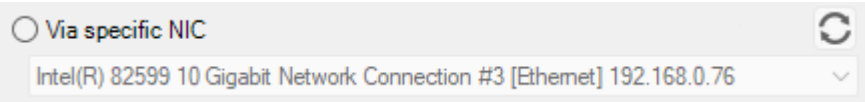
NIC Discovery Method

Via all NICs



Discovery packets are sent using all active network interfaces. This is the recommended option when the location of the radio is unknown or when multiple adapters are present.

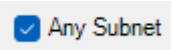
Via specific NIC



Limits discovery traffic to a single selected network interface. Use this when the radio is known to be on a specific adapter or when troubleshooting multi-NIC systems. Pressing the button will refresh the nics available to be used. Note: some interfaces are not of type Ethernet/WiFi so you may need to uncheck the advanced option, limit to Ethernet/WiFi interfaces.

Subnet

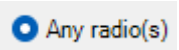
Any Subnet



Allows a radio to be included during discovery or connection even if it is not on the same subnet as the network interface being used. When disabled, radios are only included if they are on the same subnet as the NIC that they are found via..

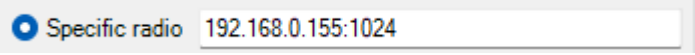
Radio Selection

Any radio



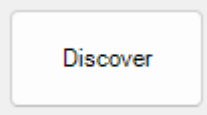
Discovery packets are broadcast in a way that allows all responding radios to be discovered. Any radio that responds will be shown when Discover is pressed. You can direct a discovery packet to a specific radio using the next option if a radio fails to respond.

Specific radio

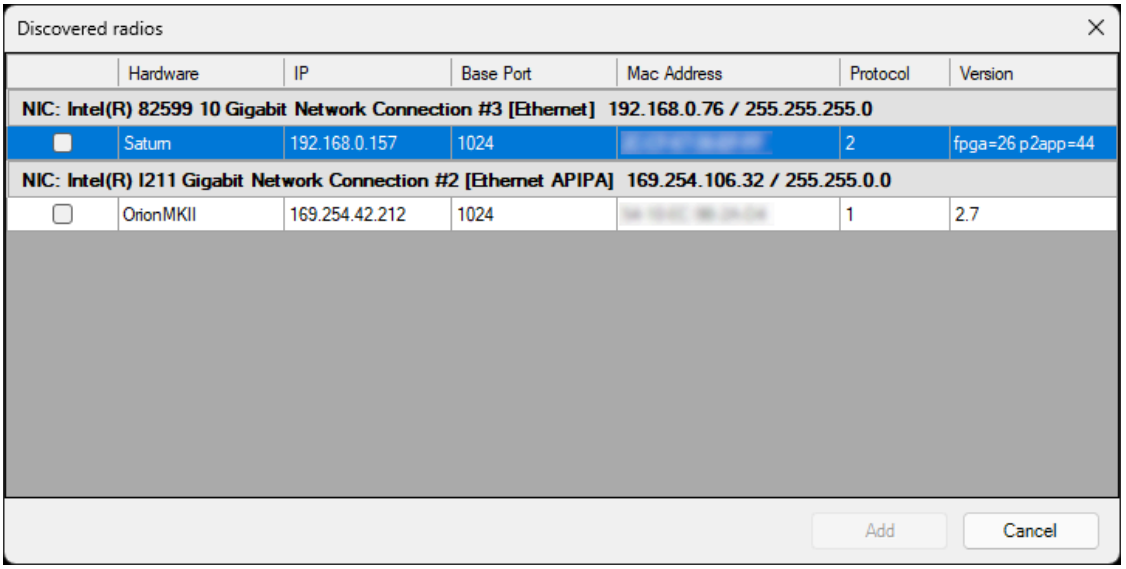


Discovery packets are sent targeting a specific IP address and port. This is used when the address of the radio is already known. Urls can also be used if you are using a dynamic dns service. Discovery messages directed to an Anan Protocol 1 radio are not always responded to (more details later).

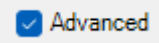
Discover Button



Pressing Discover sends discovery packets using the current settings and radio selection mode, along with some of the advanced settings. Any radios that respond will be shown and may then be added to the radio list by the user. Select a radio(s) using the checkbox on the left and then press the add button.

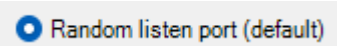


Advanced Options



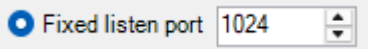
Tick this on to show/hide the advanced options. These options will be used irrespective of their visibility.

Random listen port (default)



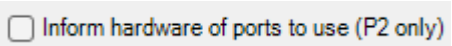
Thetis selects an available UDP port automatically for receiving discovery replies. This is the recommended option for most users.

Fixed listen port



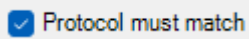
Forces Thetis to listen on a specific UDP port. This may be required when using strict firewall rules, NAT, or port forwarding. Care must be taken to avoid port conflicts. If using remote sites, then the local site running Thetis may want to port forward this port directly to the Thetis machine ip.

Inform hardware of ports to use (P2 only)



When enabled, radios are informed of their local ports to use to take data in and which local ports to use when it sends data back to Thetis to the listen port. This option only applies to Protocol-2 radios. The radio will always listen on the 1024 port for discovery packets, but the other inbound default ports 1025-1029 can be changed. Doing so will also change the local outbound ports (normally 1025-1041) the radio uses to send data back to the listen port at the Thetis end. *NOTE: this is not supported in any current P2 firmware*

Protocol must match

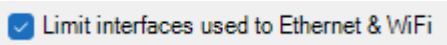


Controls protocol checking when connecting to a radio that already exists in the radio list.

Each radio stored in the list records the protocol it was discovered or added with. If this option is enabled, Thetis will only connect if the radio reports the same protocol.

If disabled, Thetis will attempt to connect regardless of the stored protocol. This is useful when radio firmware is changed frequently in a development environment and avoids the need to remove and re-add the radio.

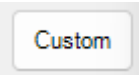
Limit interfaces used to Ethernet & WiFi



When enabled, discovery and connection traffic is limited to interfaces that identify themselves as Ethernet or Wi-Fi adapters.

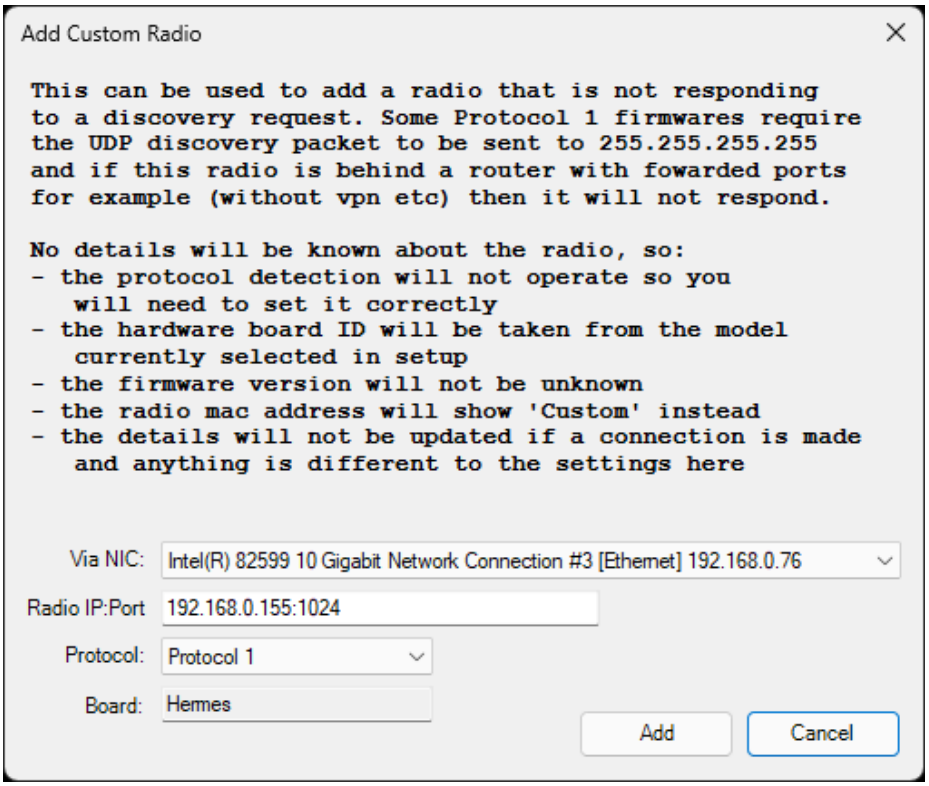
When disabled, additional interface types may be used, including virtual or specialised interfaces such as type 53.

Custom button



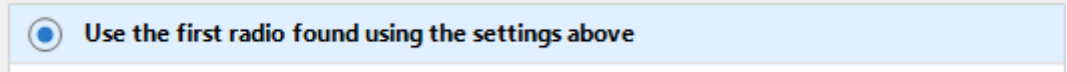
Allows a custom radio to be manually added to the radio list. This is intended for radios that do not respond to discovery packets or are otherwise unreachable via normal discovery methods. Custom radios are added directly using user-supplied details through the custom radio form.

The custom form has additional information explaining its use.



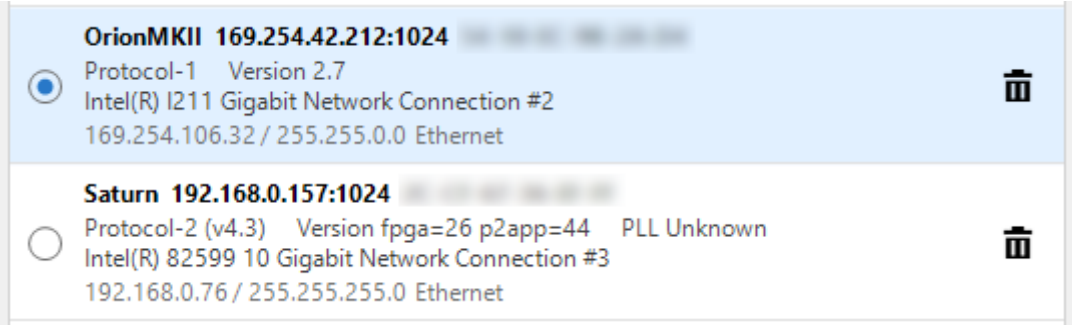
Radio List Behaviour

Use the first radio found using the settings above



The radio list always contains this special entry representing the first radio found using automatic discovery with the current network settings. When this option is in use and Thetis is powered up, Thetis will attempt discovery using the current settings and connect to the first responding radio. Details from the radio will be temporarily updated into this row, but will not persist if Thetis power is turned off. This row/option can not be removed. It should be noted that if this is used in an environment where multiple radios can be discovered the radio used is not always guaranteed to be the same each time.

Other radios may appear here that have been added by the user



Other radios appear in the list only if they are explicitly added by the user, either by running a manual Discover or by adding a Custom radio.

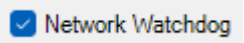
When Thetis is turned on and this radio is selected a directed discovery will be made. If the radio is found then any differences in the returned data will be updated and the connection will happen as expected.

If a custom radio is added, a discovery will not be made and Thetis will attempt to blindly connect. Time out using this method can be 5-10 seconds if the radio is not able to be contacted.

The trash icon can be clicked to remove that entry.

Other Features

Network Watchdog



Checking this on provides a timeout to each network message sent to the radio. If off, then Thetis can wait indefinitely for a reply to a message, and as such may not know that a radio has disconnected, lost the network connection, has been turned off, or if the firmware has crashed. If this option is unchecked, and something happens to the radio, Thetis will not know and may enter a frozen state waiting for a reply.

The watchdog setting is also sent to the radio firmware as part of the general packet, and it will use this to monitor its own state. Details to be confirmed. *(TODO: ask Rick n1gp)*

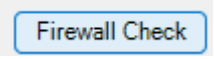
Network Throttle Index Tweak

This option will change windows network throttling behaviour. The current state will already be read from the registry and shown. It has been found that if setting this registry entry to 0xFFFFFFFF then throttling is disabled and can improve and prevent issues in Thetis data streams. Leaving this setting as default (0x0000000A) will allow Windows to throttle other network streams, such as Thetis, to maintain media playback continuity in such things as YouTube for example.

This setting can only be changed when Thetis is running as admin, and you need to reboot to have this change take effect. It is recommended that this option is checked ON.

ApacheLabs community related info : https://community.apache-labs.com/viewtopic.php?t=3787

Firewall Check



Thetis will clear and re-add firewall rules that it requires. This can only be performed when Thetis is running as admin. Thetis.exe will be permitted to use the following

UDP in/out all ports
TCP in/out all ports

With the following firewall profiles : Public, Domain and Private

The entries will be named:

Thetis Allow IN TCP
Thetis Allow OUT TCP
Thetis Allow IN UDP
Thetis Allow OUT UDP

Some remote (non vpn) configuration examples

In all cases ensure that in/out data is within your connection bandwidth limits, otherwise seq errors, or spectral/audio glitches and corruption may occur. You can use the Test tab bandwidth monitor to see what Thetis is doing, however be mindful of other data that might also be using the link.

It should be noted that for Protocol 1 radios, it may be fine to use the default listen port option in Thetis. However, setting a custom one ensures that the radios feed their data back to a known port, which could also be port mapped to the Thetis machine IP. This reverse port mapping for incoming data from the radio is more than likely required for a Protocol 2 radio.

Protocol 1 : Thetis will send to a single port, and receive from the radios single port to the single listen port

Protocol 2 :Thetis will send to 5 ports (1024-1029 default) and can receive from 17 ports on the radio (1024-1041 default) to the single listen port.

Remote Protocol 1 Anan Radio

Unfortunately, a discovery packet can not be directed to a Protocol 1 Anan, it will not respond. It only currently recognises discovery packets that are sent to 255.255.255.255 on the network segment that the radio sits on.

This prevents a radio being added using the normal discovery method. However it can still be made to work using the add Custom radio option and at the time of writing has been tested with 10e(revised), Hermes and a couple of other protocol 1 radios.

1) Configure the remote router such that on its external IP it forwards UDP packets arriving on port X to the internal IP of the radio port 1024. X can be 1024, but keeping a radio hidden is perhaps worthwhile using a different exposed port X, such as 50101 or whatever you feel like.

2) Add a custom radio in Thetis at the client end, that is of the correct board type (taken from the model selected in setup). The radioIPport should be y.y.y.y:X where y.y.y.y is your external IP of the remote site, and X is the port you exposed and forwarded in 1) above. Make sure the correct protocol is selected.

A new custom connection radio will be added, which bypasses the need for radio discovery. The list item will not be updated by any discovery information, as a discovery will not be made. Power on in Thetis will attempt to make the connection. This may take up to 10 seconds or so to time out if there is an issue.

Remote Protocol 1 HL2 Radio

Unlike the Anan’s above, the HL2 radio firmware can accept a directed discovery packet, so adding a custom radio is not normally required. A directed discovery using the specific radio ip:port should be able to be made and the radio can be added as normal using that method. You would set up the port forwarding etc. on the remote site as per the Anan example above, and press the Discover button. If the radio is found it should show up and can be added as normal.

Remote Protocol 2 Radio

Protocol 2 can respond to a directed discovery packet, which allows it to be found easily when remote. *(It should be noted that currently there is a bug, in that protocol 2 firmware does not respond to a network segment directed broadcast, for example sending a discovery packet to 192.168.1.127 when the radio is on a 255.255.255.192 subnet)*

To add a protocol 2 radio do the following:

1) Configure the remote router such that the external ip ports 1024-1029 UDP are forwarded to ports 1024-1029 on the internal network IP of the radio. You may wish to use other ports to ‘hide’ the radio, so for example you could forward 51024-51029 UDP to the internal ip of the radio on ports 1024-1029.

If the option [Inform hardware of ports to use (P2 only)] is used in Thetis and the FW supports it (none currently do) then the discovery port will stay on 1024, but if you were to use 51024 as the exposed port, then the radio would then expect data to arrive on 1024,51205-51029. As this is not currently implemented it is a level of complexity that can be ignored.

2) The radio will use its own local ports, default 1025-1041 to send data back to the client's listen port. Configure a fixed listen port, for example 4445. That port then needs to be port-forwarded on the client end router to forward all UDP traffic arriving on that listen port back to the Thetis machine IP.

3) You should now be able to perform a directed discovery, so enter the specific radio ip details:port and press the discover button. The radio should be listed and you can add it as normal.